

A Survey of Deep Learning for Extreme Weather Forecasting

InteractiveSurvey

Abstract

Extreme weather events pose escalating threats to global infrastructure and human safety, driving an urgent need for forecasting systems that surpass the computational latency and smoothing limitations of traditional Numerical Weather Prediction. Consequently, the meteorological community is increasingly adopting data-driven approaches to capture complex, non-linear atmospheric dynamics with unprecedented efficiency. This survey provides a comprehensive review of deep learning methodologies specifically tailored for extreme weather forecasting, focusing on architectures capable of modeling rare, high-impact events beyond general meteorological applications. We synthesize advancements in generative modeling, hybrid physical-data frameworks, and specialized networks designed to preserve the statistical tails of atmospheric distributions while addressing challenges like data scarcity and uncertainty quantification. The review identifies a critical shift toward transformer and graph-based global models that effectively handle spherical geometry, alongside generative approaches such as diffusion models and Generative Adversarial Networks that revolutionize probabilistic ensemble generation by preserving fine-scale structures. Furthermore, we highlight the emergence of hybrid frameworks integrating physical constraints to ensure consistency and interpretability in simulating severe phenomena like tropical cyclones. By offering a unified taxonomy and outlining a roadmap for future research, this work elucidates how combining artificial intelligence efficiency with physical fidelity is reshaping the landscape of extreme event prediction to enhance societal resilience.

1 Introduction

Extreme weather events, ranging from tropical cyclones to severe precipitation and heatwaves,

pose escalating threats to global infrastructure, economies, and human safety [1]. As climate change intensifies the frequency and magnitude of these phenomena, the demand for accurate, timely, and high-resolution forecasting has never been more critical. Traditional Numerical Weather Prediction (NWP) systems, while physically rigorous, often struggle with the computational latency required for rapid ensemble generation and frequently smooth out extreme values due to inherent numerical diffusion. Consequently, the meteorological community is increasingly turning to data-driven approaches that can leverage vast archives of reanalysis data and satellite observations to capture complex, non-linear atmospheric dynamics with unprecedented efficiency and precision.

This survey paper provides a comprehensive review of deep learning methodologies specifically tailored for extreme weather forecasting, moving beyond general meteorological applications to focus on architectures capable of modeling rare, high-impact events [2]. While previous reviews have broadly covered AI in weather prediction, there remains a distinct need to synthesize advancements in generative modeling, hybrid physical-data frameworks, and specialized networks designed to preserve the statistical tails of atmospheric distributions [1]. By examining the intersection of advanced neural architectures and probabilistic generative approaches, this work aims to elucidate how modern deep learning techniques address the unique challenges of predicting extreme phenomena, including data scarcity, spatial heterogeneity, and the necessity for reliable uncertainty quantification in operational settings.

The core of this review begins by analyzing architectural innovations that have redefined spatiotemporal modeling in meteorology. We examine the shift from traditional convolutional approaches to transformer and graph-based global models that effectively handle the non-Euclidean

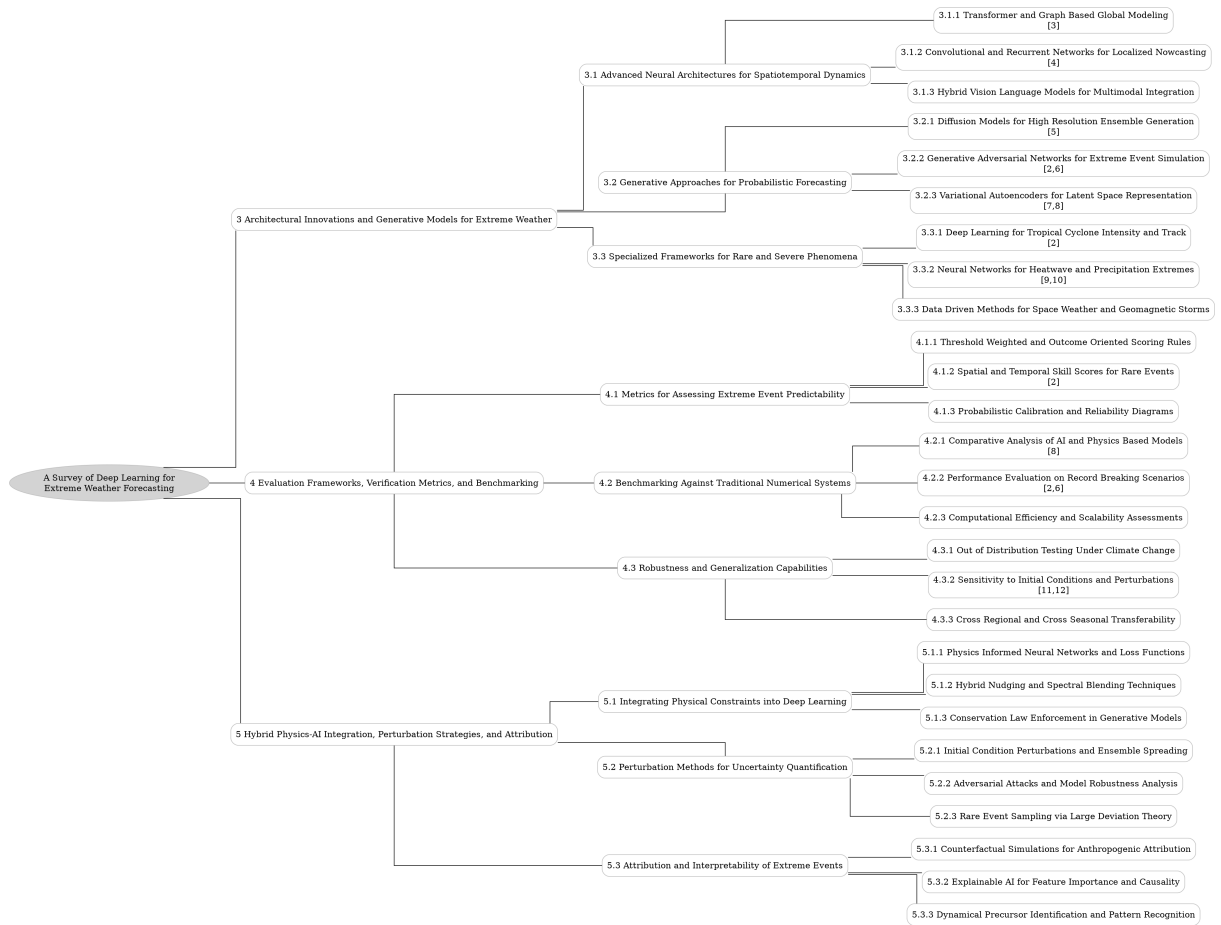


Figure 1: The outline of our survey: A Survey of Deep Learning for Extreme Weather Forecasting

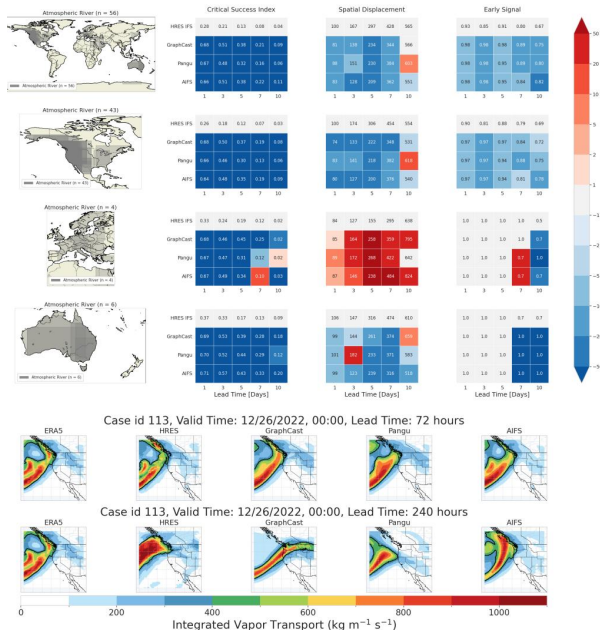


Figure 2: Chart from 'Extreme Weather Bench: A Community-Driven Benchmark Suite for High-Impact Weather Forecasting' (arXiv:2605.01126)

geometry of spherical Earth data, ensuring uniform node distribution and capturing long-range teleconnections essential for medium-range forecasting. Simultaneously, we explore localized nowcasting frameworks that integrate convolutional and recurrent networks to resolve fine-scale storm dynamics, as well as emerging hybrid vision-language models that unify spatial and semantic representations. These sections highlight how combining geometric deep learning with attention mechanisms allows for the explicit modeling of physical constraints while maintaining the scalability required for high-resolution global inference.

Subsequently, the discussion transitions to generative approaches that are revolutionizing probabilistic forecasting and ensemble generation. We detail the application of diffusion models for downscaling coarse global forecasts into kilometer-scale regional ensembles, preserving fine-scale structures often lost in deterministic predictions. The review further investigates Generative Adversarial Networks (GANs) for simulating realistic extreme events that exceed historical records, addressing the tendency of regression-based models to underestimate high-impact phenomena. Additionally, we assess Variational Autoencoders (VAEs) for their capacity to learn robust latent space representations, enabling the sys-

tematic exploration of uncertainty and the generation of diverse, physically plausible weather scenarios through stochastic formulations.

Finally, the survey dedicates significant attention to specialized frameworks designed for rare and severe phenomena, with a particular focus on tropical cyclone intensity and track prediction [2]. This section evaluates hybrid models that integrate convolutional and recurrent units to capture the non-linear evolution of cyclone trajectories, alongside physics-augmented approaches that incorporate dynamical constraints to mitigate black-box limitations. By synthesizing these diverse methodological strands, the paper bridges the gap between theoretical architectural advancements and practical applications in hazard mitigation, offering a structured perspective on how deep learning is reshaping the landscape of extreme event prediction.

The primary contributions of this survey are threefold. First, it offers a unified taxonomy of deep learning architectures specifically optimized for extreme weather, distinguishing between global modeling, localized nowcasting, and generative ensemble techniques [2]. Second, it critically analyzes the integration of physical constraints within data-driven frameworks, highlighting strategies to ensure physical consistency and interpretability in extreme event simulations. Third, by identifying current limitations such as mode collapse in generative models and data scarcity for rare events, this work outlines a clear roadmap for future research, guiding the development of next-generation hybrid models that combine the efficiency of artificial intelligence with the fidelity of physical laws to enhance societal resilience against climate extremes.

2 Architectural Innovations and Generative Models for Extreme Weather

2.1 Advanced Neural Architectures for Spatiotemporal Dynamics

2.1.1 Transformer and Graph Based Global Modeling

Transformer and graph-based architectures have emerged as dominant paradigms for global weather modeling, effectively addressing the non-Euclidean nature of spherical data. Unlike traditional convolutional approaches that struggle with polar distortions, graph neural networks (GNNs)

utilize icosahedral grids or multi-scale graph structures to represent the Earth's surface, ensuring uniform node distribution and preserving spatial continuity [3]. These models typically employ an encoder-processor-decoder framework where the processor leverages message-passing mechanisms to simulate complex atmospheric dynamics across varying spatial granularities, mimicking multigrid methods to capture interactions from local turbulence to planetary waves.

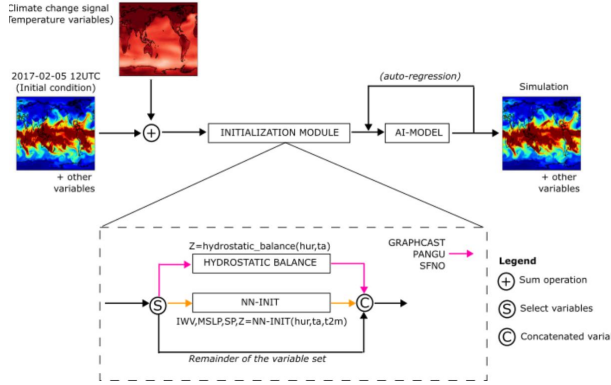


Figure 3: Chart from 'Harnessing AI data-driven global weather models for climate attribution: An analysis of the 2017 Oroville Dam extreme atmospheric river' (arXiv:2409.11605)

Complementing graph-based spatial discretization, Transformer architectures, particularly those utilizing Swin Transformer blocks with shifted windows, provide efficient long-range dependency modeling essential for medium-range forecasting. By integrating cube embedding layers to reduce spatiotemporal dimensions, these models process atmospheric variables as sequences of tokens, enabling the capture of global teleconnection patterns that purely local operators might miss. Advanced variants often combine spherical convolutions with attention mechanisms within a U-Net structure, balancing the computational efficiency of localized processing with the global receptive fields required for accurate pressure and wind field predictions over extended lead times.

The synergy between these approaches is evident in hybrid designs that embed Transformer layers within graph processors or utilize graph-based encoders to feed vision Transformers. Such configurations allow for the explicit modeling of physical constraints while maintaining the scalability needed for high-resolution global inference. By optimizing auto-regressive rollout strategies and incorporating earth-specific region refinement,

these models achieve superior skill in predicting extreme events and maintaining stability over multi-day forecasts. Consequently, the integration of geometric deep learning with attention-based global modeling represents a significant shift toward data-driven systems that rival conventional numerical weather prediction in both accuracy and computational speed.

2.1.2 Convolutional and Recurrent Networks for Localized Nowcasting

Convolutional Neural Networks (CNNs) have become foundational for localized nowcasting by effectively exploiting spatial context within meteorological fields. Architectures such as U-Net employ encoder-decoder structures with skip connections to preserve high-resolution features while capturing multi-scale dependencies, proving essential for predicting extreme precipitation and wind gusts where deterministic models often smooth out extrema. By integrating transpose convolutions for upsampling and concatenating encoding path features, these models reconstruct fine-scale variability from coarse inputs, significantly enhancing the representation of local storm dynamics and surface temperature anomalies.

To address the temporal evolution of weather systems, Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) variants, are frequently coupled with convolutional layers in frameworks like ConvLSTM [4]. These hybrid architectures mitigate the vanishing gradient problem inherent in standard RNNs, allowing the model to retain relevant historical information over extended sequences. The convolutional components extract spatial patterns from radar or satellite imagery, while the recurrent units analyze sequential aspects to forecast the motion and transformation of precipitation systems, effectively modeling the non-linear advection of atmospheric variables over short timescales.

$$RMSE = \sqrt{\frac{1}{N} \sum_{n=1}^N \|X - X^{pred}\|^2},$$

Figure 4: Chart from 'ACCESSING CONVECTIVE HAZARDS FREQUENCY SHIFT WITH CLIMATE CHANGE USING PHYSICS - INFORMED MACHINE LEARNING' (arXiv:2310.03180)

Recent advancements combine these spatial and temporal mechanisms into sophisticated spatiotemporal predictors that outperform traditional extrapolation methods. Models like Nowcast-Net and DGMR utilize deep generative approaches to produce probabilistic forecasts that maintain physical consistency and sharpness in extreme events. Furthermore, integrating attention mechanisms or adaptive Fourier neural operators within these convolutional-recurrent frameworks enables the capture of long-range dependencies and frequency-domain characteristics. This synergy allows for accurate, high-resolution predictions of localized phenomena, bridging the gap between global reanalysis data and regional impact forecasting.

2.1.3 Hybrid Vision Language Models for Multimodal Integration

Hybrid Vision Language Models (VLMs) for multimodal integration increasingly leverage transformer architectures to unify spatial and semantic representations. By employing self-attention mechanisms, these models capture long-range dependencies and teleconnections essential for complex spatiotemporal tasks, overcoming the locality constraints inherent in traditional Convolutional Neural Networks. The integration of embedding layers transforms heterogeneous inputs, such as meteorological fields or image patches, into a shared vector space, facilitating the joint processing of cross-spatial dynamics that significantly enhance prediction performance compared to independent field analysis.

To address the computational demands of high-resolution data, recent hybrid designs incorporate hierarchical structures like shifted windows and dilated convolutions to achieve fully receptive fields without prohibitive cost. These architectures often combine encoder-decoder frameworks with multi-head attention mechanisms, allowing for precise aggregation of node-specific information while dynamically adjusting weights based on contextual relevance. This synergy enables the model to effectively disentangle physical dynamics from unknown factors, ensuring that intricate variations in temporal dynamics are captured alongside static spatial features.

Furthermore, advanced hybrid models integrate physical constraints, such as conservation laws and causal consistency regularization, directly into the learning objective to improve robustness in

extreme event modeling. By utilizing hybrid loss functions and leveraging automatic differentiation, these systems enforce physical plausibility while maintaining the computational efficiency required for large-ensemble forecasting. The resulting frameworks not only accelerate inference speeds by orders of magnitude compared to numerical weather prediction models but also provide interpretable insights into spatio-temporal interaction landscapes through automated concept summarization.

2.2 Generative Approaches for Probabilistic Forecasting

2.2.1 Diffusion Models for High Resolution Ensemble Generation

Diffusion models have emerged as a transformative approach for generating high-resolution ensemble forecasts, addressing the computational bottlenecks of traditional numerical weather prediction [5]. By leveraging a forward noising process and a learned reverse denoising trajectory, these architectures effectively sample from complex, multi-modal probability distributions inherent in atmospheric dynamics. Unlike deterministic deep learning models that often suffer from blurring effects due to mean-squared error optimization, diffusion frameworks preserve fine-scale spatial structures and extreme event statistics, making them particularly suitable for capturing the uncertainty required in ensemble forecasting.

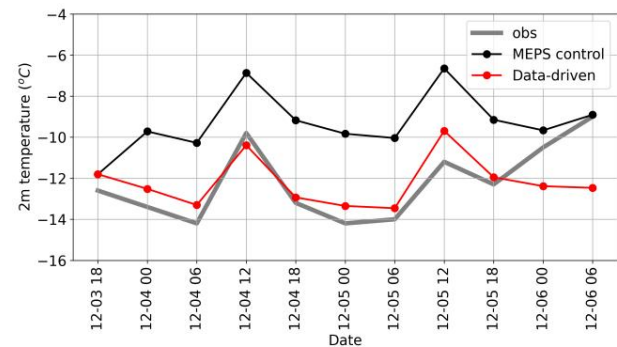


Figure 5: Chart from 'REGIONAL DATA-DRIVEN WEATHER MODELING WITH A GLOBAL STRETCHED-GRID' (arXiv:2409.02891)

Recent advancements focus on conditional generation techniques to downscale coarse global forecasts into kilometer-scale regional ensembles. Methods such as Anemoi-D $\mathcal{S}$ and residual diffusion models learn the conditional distribution of high-frequency residuals, defined as the differ-

ence between high-resolution ground truth and interpolated low-resolution inputs. This strategy allows the model to inject physically consistent small-scale variability—such as convective structures and sharp thermal gradients—into the forecast fields without re-running expensive dynamical cores. Consequently, these systems can generate large ensemble members efficiently, providing probabilistic insights into localized phenomena like tropical cyclone intensity and precipitation extremes that are often smoothed out in lower-resolution simulations.

Furthermore, the integration of diffusion processes with graph neural operators and adaptive Fourier layers enhances the temporal consistency and global coherence of generated ensembles. By training on historical reanalysis data and high-resolution limited-area models, these hybrid architectures emulate the spread-skill relationship of physics-driven ensembles while significantly reducing inference latency. The ability to generate diverse, high-fidelity scenarios from a single initial condition enables robust uncertainty quantification for medium-range forecasting. As these models evolve, they promise to bridge the gap between data-driven efficiency and the physical fidelity required for operational high-resolution ensemble prediction systems.

2.2.2 Generative Adversarial Networks for Extreme Event Simulation

Generative Adversarial Networks (GANs) have emerged as a pivotal architecture for simulating extreme weather events, addressing the inherent tendency of regression-based deep learning models to produce overly smooth forecasts that underestimate high-impact phenomena [2]. By pitting a generator against a discriminator, GANs learn to reconstruct the complex, non-Gaussian probability distributions characteristic of rare events such as tropical cyclones and severe precipitation. This adversarial training framework effectively captures high-frequency spatial details and sharp gradients often lost in traditional mean-squared-error optimization, enabling the generation of realistic, high-resolution synthetic scenarios that preserve the statistical properties of extreme tails.

To overcome the scarcity of observational data for rare occurrences, GANs are increasingly integrated with advanced sampling strategies and physical constraints. Techniques such as transfer learning allow models trained on frequent, less

extreme events to generalize toward rarer intensities, while nested extreme event structures leverage the hierarchical nature of atmospheric disturbances. Furthermore, recent advancements incorporate conservation laws and symmetries directly into the loss functions or network architecture, ensuring that generated "gray swan" events—plausible but unobserved extremes—remain physically consistent. This hybrid approach mitigates the risk of generating artifacts and enhances the reliability of synthetic datasets used for risk assessment and infrastructure design.

Despite their promise, applying GANs to operational meteorology presents challenges regarding mode collapse and the computational cost of training stable models on global scales. Current research focuses on refining architectures, such as combining convolutional backbones with vision transformers, to improve the fidelity of long-lead forecasts. Additionally, efforts are underway to utilize GANs for downscaling coarse global predictions to hazard-relevant local scales, particularly for wind gusts and convective precipitation. As these models evolve, they offer a robust pathway for augmenting ensemble forecasting systems, providing critical insights into the dynamics of extreme events that exceed the limits of historical records [6].

$$\text{RMSE} = \sqrt{\frac{1}{NT} \sum_{n=1}^N \sum_{t=1}^T \left(y_{nt} - \frac{1}{M} \sum_{m=1}^M \hat{y}_{nt}^{(m)} \right)^2}$$

Figure 6: Chart from 'On the Predictive Skill of Artificial Intelligence-based Weather Models for Extreme Events using Uncertainty Quantification' (arXiv:2511.17176)

2.2.3 Variational Autoencoders for Latent Space Representation

Variational Autoencoders (VAEs) have emerged as a pivotal architecture for learning robust latent space representations in meteorological modeling, addressing the limitations of deterministic autoencoders by enforcing probabilistic constraints. Unlike standard encoders that map inputs to fixed points, VAEs parameterize the latent space as a distribution, typically Gaussian, characterized by mean and variance vectors [7]. This stochastic formulation allows the model to capture the inherent uncertainty in atmospheric dynamics, facilitating the generation of diverse and physically plausible

weather scenarios rather than single-point predictions that may suffer from blurring or mode collapse.

$$\Psi_{ij} = \sum_{k=0}^{K-1} \Psi_{kij},$$

Figure 7: Chart from 'A REVIEW OF RADAR-BASED NOWCASTING OF PRECIPITATION AND APPLICABLE MACHINE LEARNING TECHNIQUES' (arXiv:2005.04988)

The optimization of these models relies on maximizing the Evidence Lower Bound (ELBO), which balances reconstruction fidelity against the Kullback-Leibler divergence between the learned posterior and a predefined prior. This regularization ensures that the latent manifold remains continuous and smooth, enabling effective interpolation between distinct weather states. In practice, this is often implemented within encoder-decoder frameworks, such as UNet variants, where skip connections preserve high-frequency spatial details while the variational bottleneck compresses global contextual information into a compact, semantically rich feature space suitable for downstream forecasting tasks.

Furthermore, the integration of VAEs with advanced architectural components, such as normalizing flows or spectral operators, enhances their capacity to model complex, non-Gaussian distributions inherent in geophysical data. By decoupling latent variables from specific observational noise, these models facilitate disentangled representations where distinct factors of variation correspond to interpretable physical phenomena. This capability is critical for generative downscaling and long-term climate projection, as it allows for the systematic exploration of extreme events and the quantification of predictive uncertainty within a unified deep learning framework [8].

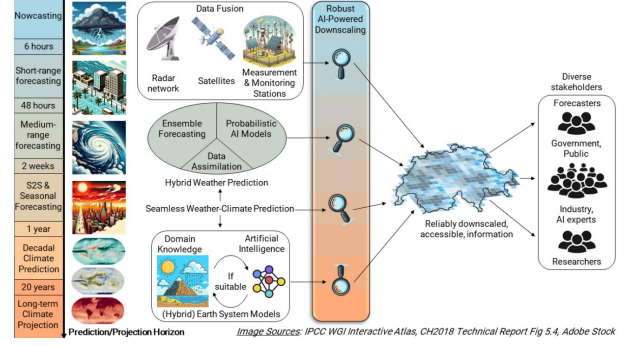


Figure 8: Chart from 'Next-Generation Earth System Models: Towards Reliable Hybrid Models for Weather and Climate Applications' (arXiv:2311.13691)

2.3 Specialized Frameworks for Rare and Severe Phenomena

2.3.1 Deep Learning for Tropical Cyclone Intensity and Track

Deep learning architectures have increasingly supplanted traditional statistical methods for predicting tropical cyclone (TC) tracks and intensity by exploiting complex spatiotemporal patterns in heterogeneous datasets [2]. Hybrid models integrating Convolutional Neural Networks (CNNs) with Gated Recurrent Units (GRUs) effectively capture the non-linear evolution of cyclone trajectories, leveraging multi-dimensional feature selection to reduce forecast errors in cyclone-eye localization [2]. Furthermore, physics-augmented approaches enhance these data-driven frameworks by incorporating dynamical constraints, allowing for more realistic reconstructions of wind and pressure fields from sparse satellite imagery while mitigating the black-box limitations inherent in pure neural network formulations.

Despite these advancements, accurate intensity forecasting remains challenging due to the scarcity of high-resolution training data for extreme events and systematic underestimations within global re-analysis products like ERA5. To address data paucity, researchers employ sophisticated data augmentation techniques and concept-gated fine-tuning frameworks that prevent overfitting while improving the representation of low-pressure tails and high-wind extremes. Ensemble deep learning strategies further refine predictions by aggregating outputs from diverse models, thereby quantifying uncertainty and enhancing robustness against the chaotic nature of rapid intensification processes that often elude single-model deterministic forecasts.

Recent developments focus on coupling deep learning emulators with numerical weather prediction systems to optimize initial condition perturbations and improve ensemble spread. Generative frameworks now integrate physical conservation laws to correct biases in global models, ensuring that predicted storm structures adhere to fundamental atmospheric dynamics. While current AI-driven systems demonstrate superior computational efficiency and competitive skill in medium-range track forecasting compared to operational baselines, ongoing research prioritizes resolving resolution discrepancies and enhancing the fidelity of inner-core intensity estimates to support reliable disaster mitigation strategies.

2.3.2 Neural Networks for Heatwave and Precipitation Extremes

Neural networks have emerged as potent tools for forecasting heatwave extremes by leveraging vast climatological datasets to identify non-linear atmospheric patterns. Convolutional Neural Networks (CNNs) and Graph Neural Networks (GNNs) are frequently employed to process surface temperature and 500 hPa geopotential height fields, enabling the prediction of long-lasting heat events up to fifteen days in advance. These architectures effectively overcome traditional correlation time limits by learning from millennial-scale simulated climate data, often utilizing transfer learning and class undersampling techniques to address the rarity of extreme occurrences while maintaining high regional accuracy.

For precipitation extremes, deep learning approaches focus on capturing complex spatiotemporal dynamics inherent in convective systems and heavy rainfall events [9]. Models such as NowcastNet and DGMR utilize deep generative frameworks and radar data to excel in short-term nowcasting, whereas adaptive Fourier neural operators in architectures like FourCastNet facilitate high-resolution global forecasting [10]. By integrating variables such as humidity and wind components, these networks simulate the evolution of physical dynamics more efficiently than traditional numerical methods, providing probabilistic forecasts that are critical for issuing timely severe weather warnings and mitigating flood risks.

Recent advancements integrate these specific extreme weather capabilities into broader Neural Weather Models (NeWMs) that operate on global grids with significantly reduced computa-

$$L_{adv} = -\frac{1}{|X|} \sum_{\{x_{t+k\Delta t}\}_{k=0}^T \in X} \frac{1}{T} \sum_{k=1}^T \log(D(\hat{x}_{t+(k-1)\Delta t}, \hat{x}_{t+\Delta t})).$$

Figure 9: Chart from 'Kunyu: A High-Performing Global Weather Model Beyond Regression Losses' (arXiv:2312.08264)

tional costs compared to coupled physical equation solvers. The incorporation of Vision Transformers and 3D neural networks allows for the simultaneous modeling of multi-variable atmospheric states, enhancing the representation of extreme value distributions. Furthermore, the application of gradient-based optimization within these differentiable frameworks offers new avenues for sensitivity analysis and optimal perturbation growth studies, potentially transforming the understanding of predictability limits for catastrophic heat and precipitation events.

2.3.3 Data Driven Methods for Space Weather and Geomagnetic Storms

Data-driven methodologies have recently expanded from terrestrial meteorology into the domain of space weather, leveraging deep learning architectures to model the complex, non-linear dynamics of the Solar-Magnetosphere-Ionosphere system. By constructing spatio-temporal causal graphs from multi-source observational datasets spanning decades, these models identify latent relationships between solar wind parameters and geomagnetic indices that traditional physics-based simulations often miss due to computational constraints. The integration of historical storm metadata with real-time solar observations enables the reconstruction of panoramic space weather views, facilitating the detection of precursor signals for extreme geomagnetic events with improved lead times.

Advanced neural networks, including convolutional and graph-based architectures adapted for spherical geometries, are increasingly employed to forecast geomagnetic storm evolution and intensity. Unlike standard regression approaches that tend to produce overly smoothed predictions and underestimate extreme values, recent innovations incorporate adversarial training and diffusion models to preserve the granularity of storm structures. These techniques address the inherent smoothing bias in recurrent models, ensuring that rapid fluctuations in geomagnetic activity and large-scale displacement trends are accurately captured, which

is critical for predicting the onset and severity of space weather hazards.

Despite promising advancements in capturing global statistical properties, challenges remain in representing fine-grained storm characteristics and handling the non-stationary nature of space plasma environments. Current data-driven frameworks demonstrate superior performance over linear models and traditional statistical methods in predicting storm trajectories and intensity increments, yet they require robust ensemble post-processing to quantify uncertainty effectively. Future developments aim to hybridize these data-centric approaches with physical constraints, creating emulators that maintain computational efficiency while adhering to the fundamental conservation laws governing magnetospheric dynamics, thereby enhancing operational reliability for early warning systems.

3 Evaluation Frameworks, Verification Metrics, and Benchmarking

3.1 Metrics for Assessing Extreme Event Predictability

3.1.1 Threshold Weighted and Outcome Oriented Scoring Rules

Threshold-weighted scoring rules address the inherent limitation of standard proper scoring rules, which often prioritize overall distributional fit at the expense of extreme event accuracy. By applying a weight function to the score based on the magnitude of the observation or prediction, these metrics explicitly penalize errors in the tails of the distribution more severely than those in the central regime. This approach is critical for weather forecasting, where the societal impact of missing a high-intensity precipitation event or underestimating wind speeds far outweighs minor deviations in moderate conditions. Consequently, the optimization landscape shifts to encourage models to learn features specifically indicative of rare, high-impact phenomena rather than merely converging to the climatological mean.

Outcome-oriented scoring rules further refine evaluation by focusing on the utility of forecasts for specific decision-making thresholds rather than continuous probabilistic fidelity. These metrics, such as the Critical Success Index (CSI) and the Peirce Skill Score (PSS), are derived from contingency tables that bin predictions and observations into binary events based on predefined intensity

thresholds. Unlike continuous metrics like RMSE or MAE, outcome-oriented scores directly quantify the trade-off between hits, false alarms, and misses, providing a clearer picture of a model's operational reliability for warning systems. The selection of the probability threshold becomes a pivotal hyperparameter, as it dictates the balance between precision and recall, often requiring calibration against specific user risk tolerances.

Integrating these approaches involves optimizing loss functions that mirror the desired evaluation metrics, such as using weighted binary cross-entropy to approximate the maximization of the True Skill Statistic. Recent advancements leverage these principles to train deep learning models that maintain high fidelity in general conditions while demonstrating superior discrimination capability for extremes. By aligning the training objective with threshold-weighted and outcome-oriented criteria, models can overcome the tendency to underestimate extreme percentiles, ensuring that forecast skill is preserved even as lead times increase and uncertainty grows. This alignment ensures that the resulting probabilistic forecasts are not only statistically consistent but also actionable for high-stakes applications.

3.1.2 Spatial and Temporal Skill Scores for Rare Events

Evaluating forecast skill for rare extreme events necessitates metrics that remain equitable under severe class imbalance, where standard accuracy measures often fail due to the dominance of non-event samples. Consequently, skill scores independent of event frequency, such as the Peirce Skill Score (also known as the True Skill Statistic) and the Heidke Skill Score, are prioritized to distinguish genuine predictive capability from random chance. These equitable metrics assign greater weight to the correct identification of rare occurrences, thereby discouraging models from converging on trivial solutions that simply predict the climatological mean or the most frequent class.

To address the spatial displacement errors inherent in high-resolution forecasting of localized phenomena, the Fractional Skill Score is employed across varying neighborhood scales, typically ranging from one to thirty kilometers. This approach relaxes the requirement for exact pixel-to-pixel matching, acknowledging that a forecast retaining the correct structure and intensity but slightly shifted in space still possesses significant

utility. By aggregating hits within defined spatial windows, this metric provides a more robust assessment of a model’s ability to capture fine-scale features like convective storms or typhoon tracks without being overly penalized for minor localization inaccuracies.

Temporal evaluation further complements spatial analysis by examining skill degradation over increasing lead times, often utilizing the Critical Success Index and Probability of Detection at specific intensity thresholds. Studies indicate that while deep learning architectures may outperform traditional numerical weather prediction systems in the short term, their skill scores for extreme precipitation and wind speeds tend to decay as the forecast horizon extends [2]. Therefore, comprehensive benchmarking involves calculating these categorical scores across multiple temporal intervals and precipitation rates to ensure models maintain reliability for high-impact decision-making throughout the relevant forecast window.

3.1.3 Probabilistic Calibration and Reliability Diagrams

Probabilistic calibration is essential for transforming raw neural network outputs into trustworthy uncertainty estimates, particularly in extreme weather scenarios where deterministic biases can severely compromise decision-making. Unlike point-wise metrics such as Mean Absolute Error, which often encourage models to predict conditional means and suppress variance, calibration ensures that predicted probabilities align empirically with observed frequencies. This alignment is critical for applications like flash flood warning systems, where the reliability of tail events dictates operational safety and resource allocation.

Reliability diagrams serve as the primary diagnostic tool for assessing this alignment by plotting the observed frequency of an event against the forecast probability across discrete bins. In a perfectly calibrated system, the curve follows the diagonal line of identity; deviations indicate systematic miscalibration, such as overconfidence or underconfidence. For ensemble-based approaches, rank histograms provide a complementary assessment of ensemble spread, where a uniform distribution signifies that the observations are statistically indistinguishable from the ensemble members, whereas U-shaped or dome-shaped profiles reveal insufficient diversity or excessive spread, respectively.

Recent advancements integrate these diagnostics directly into the training loop through proper scoring rules like the Continuous Ranked Probability Score (CRPS) and Prediction Interval Coverage Probability (PICP). Modern generative frameworks, including diffusion models and conditional quantile regression, leverage these metrics to refine the full predictive distribution rather than optimizing solely for pixel-wise accuracy. By minimizing the divergence between forecasted and observed cumulative distribution functions, these methods produce well-calibrated ensembles that maintain structural fidelity while accurately quantifying the inherent stochasticity of atmospheric processes.

3.2 Benchmarking Against Traditional Numerical Systems

3.2.1 Comparative Analysis of AI and Physics Based Models

Purely data-driven AI models excel at capturing complex nonlinear patterns from historical observations, often surpassing traditional Numerical Weather Prediction (NWP) systems in short-term forecasting speed and accuracy for specific variables [8]. However, these black-box approaches frequently struggle with generalization under non-stationary climate conditions, as they lack explicit constraints to enforce conservation laws or thermodynamic consistency. Consequently, while deep learning architectures like Transformers and Neural Operators achieve rapid inference, they remain prone to generating physically implausible states, particularly when extrapolating beyond the distribution of training data or predicting rare extreme events.

In contrast, physics-based models rely on the iterative integration of partial differential equations, ensuring strict adherence to fundamental physical principles such as mass and energy conservation. This structural rigidity grants them superior robustness in long-lead predictions and changing climate regimes, albeit at a prohibitive computational cost that limits ensemble sizing and spatial resolution. The primary limitation of purely dynamical approaches lies in their susceptibility to error accumulation over time and the difficulty in parameterizing sub-grid scale processes, which can degrade forecast fidelity despite their theoretical soundness.

Recent advancements increasingly favor hybrid

frameworks that integrate physical constraints directly into neural network architectures, aiming to synthesize the efficiency of AI with the reliability of dynamical systems. By embedding differential equations as soft constraints within loss functions or hard constraints within network layers, physics-informed machine learning models mitigate the risk of unphysical predictions while accelerating convergence [8]. This synergistic approach not only enhances interpretability by aligning latent representations with known meteorological mechanisms but also improves the detection of extreme events, offering a viable pathway toward operational systems that are both computationally efficient and physically consistent.

3.2.2 Performance Evaluation on Record Breaking Scenarios

Evaluating model performance on record-breaking scenarios necessitates rigorous out-of-sample testing protocols that isolate extreme events from standard climatological distributions. Standard temporal splits often fail to capture the rarity of such phenomena; therefore, specialized datasets like HR-Extreme-V2 are constructed to focus exclusively on high-impact occurrences across extended periods. By cropping feature maps to specific event regions and expanding temporal coverage to include recent anomalies, these benchmarks ensure that models are assessed on their ability to generalize to unseen intensities rather than merely interpolating historical patterns.

Traditional accuracy metrics prove insufficient for these tasks due to severe class imbalance, where correct predictions of common weather states dominate the score. Consequently, evaluation frameworks prioritize equitable skill scores, precision, recall, and the Critical Success Index (CSI) at high thresholds to penalize missed detections of rare events heavily. This approach mitigates the tendency of models to under-predict extremes or over-smooth forecasts, ensuring that performance metrics reflect the actual utility of the system in life-saving applications where identifying the tail end of the distribution is paramount.

Comparative analyses reveal that while deep learning models often surpass numerical weather prediction baselines in short-term nowcasting of extreme thresholds, performance typically degrades as lead times increase [2]. Advanced architectures demonstrate robustness by maintaining superior median scores across multiple extreme cate-

gories compared to advection methods and operational systems, even when facing data scarcity. However, the persistence of systematic biases in predicting record-shattering values highlights the ongoing challenge of capturing non-stationary climate behaviors, necessitating future work on probabilistic forecasting and ensemble methods to better quantify uncertainty in these critical scenarios [6].

3.2.3 Computational Efficiency and Scalability Assessments

Computational efficiency assessments reveal a paradigm shift where data-driven models drastically outperform traditional Numerical Weather Prediction (NWP) systems in inference speed and energy consumption. While conventional approaches like the IFS require thousands of CPU nodes and over an hour to generate ensemble forecasts, modern architectures such as FourCastNet and GraphCast achieve comparable or superior results on single GPU accelerators within seconds. This orders-of-magnitude reduction in latency enables rapid generation of large ensembles, facilitating probabilistic forecasting workflows that were previously computationally prohibitive for operational centers.

Scalability analysis highlights a critical divergence in how computational costs grow with increasing spatial resolution. Traditional NWP methods exhibit super-linear scaling, where doubling resolution often necessitates an eight-fold increase in computational resources due to stricter Courant-Friedrichs-Lewy conditions and complex dynamical cores. In contrast, deep learning models generally demonstrate more favorable scaling properties; although higher resolutions demand increased memory and floating-point operations, the architectural flexibility of transformers and convolutional networks allows for efficient parallelization on specialized hardware, avoiding the severe bottlenecks inherent in physics-based solvers.

Despite these advantages, achieving extreme high-resolution forecasting remains constrained by GPU memory capacity and the quadratic complexity of attention mechanisms in certain backbone architectures. Strategies to mitigate these limitations include employing mixed-precision arithmetic, optimizing network depth versus width trade-offs, and utilizing model parallelism across multiple accelerators. Furthermore, while training these large-scale models incurs significant upfront

carbon costs, their exceptional inference efficiency ensures that long-term operational deployment offers a sustainable alternative to resource-intensive legacy systems, provided that hardware utilization is maximized during production runs.

3.3 Robustness and Generalization Capabilities

3.3.1 Out of Distribution Testing Under Climate Change

Evaluating data-driven weather models under climate change necessitates rigorous out-of-distribution testing, as training distributions derived from historical reanalysis often fail to capture the statistical properties of future warming scenarios. Standard architectures, while proficient in medium-range forecasting within stationary climates, exhibit significant degradation when confronted with the non-stationary dynamics characteristic of extreme climate shifts. Consequently, reliance on historical baselines alone is insufficient for assessing model robustness against unprecedented extreme events, such as intensified precipitation tails or altered cyclonic frequencies, which are projected to dominate future climate regimes.

To address this distributional shift, emerging methodologies propose initializing neural forecasters with outputs from global climate models rather than traditional reanalysis products. This hybrid approach enables the systematic evaluation of model performance across diverse warming trajectories, effectively stress-testing architectures against synthetic futures that deviate markedly from observed history. By generating large-member ensembles efficiently, these techniques facilitate probabilistic assessments of rare events, allowing researchers to quantify uncertainty and reliability in tail-risk scenarios where traditional numerical weather prediction systems are computationally prohibitive.

Despite these advances, substantial challenges remain in validating model behavior under truly novel physical conditions where ground truth is inherently unavailable. Current efforts focus on analyzing regime shifts in feature attribution and error characteristics during simulated extreme events, such as heat domes and Arctic outbreaks, to identify potential failure modes before they manifest in reality. Future research must prioritize the development of physics-informed constraints and adaptive learning frameworks that maintain con-

sistency with thermodynamic laws under changing boundary conditions, ensuring that data-driven tools remain reliable for long-term climate adaptation and risk mitigation strategies.

3.3.2 Sensitivity to Initial Conditions and Perturbations

The accuracy of AI-driven weather forecasting models is intrinsically linked to the fidelity of initial conditions, which are typically derived from complex data assimilation processes integrating heterogeneous observational sources [11]. Despite advancements in assimilation techniques, inherent errors persist due to sparse observations and model biases, rendering the prediction task a sensitive initial value problem. In modern deep learning architectures, such as Fourier Neural Operators, the handling of these conditions is often optimized by folding multiple initial states into the batch dimension, enabling efficient parallel inference across large ensembles on GPU hardware. However, the propagation of these initial uncertainties through non-linear neural operators remains a critical area of investigation, as small deviations can amplify rapidly over lead time.

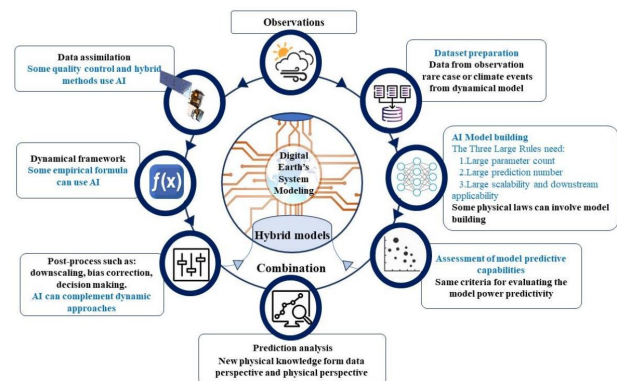


Figure 10: Chart from 'Improving Global Weather and Ocean Wave Forecast with Large Artificial Intelligence Models' (arXiv:2401.16669)

To quantify this sensitivity, researchers employ perturbation schemes that introduce controlled noise into the initial atmospheric state, effectively generating ensemble members to assess forecast stability [12]. A common methodology involves superimposing random Perlin noise vectors onto the baseline initial state, scaled by a coefficient to modulate amplitude, thereby simulating realistic analysis errors without violating physical bounds. By feeding these perturbed states into trained models, one can compute statistical metrics such as the Anomaly Correlation Coefficient and Root Mean

Square Error averaged over numerous initial conditions. These evaluations often reveal that while AI models demonstrate robustness to minor perturbations in short-term forecasts, their performance degrades distinctively compared to traditional numerical weather prediction systems as the temporal horizon extends [11].

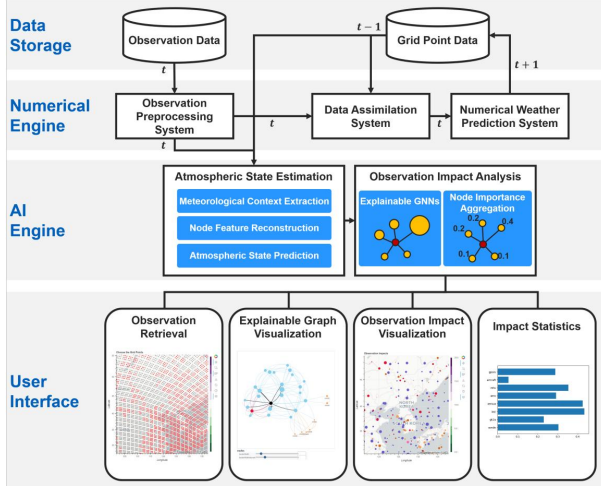


Figure 11: Chart from 'CLOUDNINE: ANALYZING METEOROLOGICAL OBSERVATION IMPACT ON WEATHER PREDICTION USING EXPLAINABLE GRAPH NEURAL NETWORKS' (arXiv:2402.14861)

Furthermore, the design of perturbation strategies must align with both the architectural characteristics of the neural network and underlying physical laws to avoid unphysical artifacts like mode collapse or significant underestimation of extreme events. While simple stochastic noise provides a baseline for uncertainty quantification, there is a growing consensus that more sophisticated methods, such as those based on singular vectors or physics-informed constraints, are necessary to capture the true growth of errors in chaotic systems. Current limitations indicate that purely data-driven approaches may struggle to resolve the nuanced dynamics introduced during the diffusion or inference process, highlighting the need for hybrid frameworks that integrate meteorological knowledge directly into the perturbation generation mechanism to enhance ensemble reliability.

3.3.3 Cross Regional and Cross Seasonal Transferability

Cross-regional transferability remains a pivotal challenge in deploying global weather models to localized contexts, particularly when shifting from training domains like the contiguous United States

to distinct climatic zones such as the Mediterranean or open oceans. While global reanalysis datasets provide a foundational baseline, models often exhibit degraded performance when applied to regions with differing topographical constraints or land-sea contrasts without explicit fine-tuning. Recent analyses indicate that neural network hybrid approaches can maintain competitive skill scores across diverse geographies; however, sensitivity to regional atmospheric regimes suggests that static global weights may fail to capture sub-regional variability essential for high-resolution forecasting.

Seasonal non-stationarity further complicates model generalization, as atmospheric dynamics undergo significant transitions between climatic periods, such as the spring turnover or summer monsoon phases. Empirical evaluations reveal that forecast skill, measured by metrics like the Peirce Skill Score, often fluctuates markedly throughout the year, with specific architectures demonstrating reduced accuracy during summer months in certain regions while performing robustly in winter. This temporal instability underscores the limitation of training on aggregated annual data, which may obscure critical seasonal patterns and lead to biased predictions during extreme weather events that are seasonally dependent.

To mitigate these transferability gaps, advanced strategies including instance normalization and physics-informed teleconnection mechanisms are being integrated to align feature distributions across different spatiotemporal domains. By leveraging temporal statistics from specific periods or employing meta-transfer learning techniques, models can better adapt to shifting climate regimes and reduce vulnerability to distributional shifts between training and inference environments. Future research directions emphasize the development of dynamic adaptation frameworks capable of adjusting to both regional heterogeneity and seasonal evolution, ensuring consistent forecast reliability across varying initial conditions and lead times.

4 Hybrid Physics-AI Integration, Perturbation Strategies, and Attribution

4.1 Integrating Physical Constraints into Deep Learning

4.1.1 Physics Informed Neural Networks and Loss Functions

Physics-Informed Neural Networks (PINNs) address the tendency of purely data-driven AI weather prediction models to generate physically inconsistent states, such as violations of mass conservation or unrealistic thermodynamic profiles. By embedding governing partial differential equations directly into the network architecture or optimization objective, these models enforce adherence to fundamental laws like the Navier-Stokes equations and hydrostatic balance. This integration is critical for maintaining stability over long autoregressive rollouts, preventing the accumulation of errors that lead to non-physical artifacts like spurious warm core structures or unbounded energy growth in tropical cyclone simulations.

The formulation of loss functions in this domain has evolved beyond standard mean squared error metrics, which often prioritize bulk distribution accuracy at the expense of extreme event fidelity. Modern approaches incorporate soft constraints where residuals of physical equations act as regularization terms, penalizing deviations from known dynamical relationships without requiring explicit numerical solvers during inference. Furthermore, specialized loss components targeting high-order moments and tail distributions are increasingly employed to mitigate amplitude damping, ensuring that the model captures the intensity and variability of high-impact weather phenomena rather than converging toward climatological means.

Advanced architectures, such as Spherical Fourier Neural Operators, further enhance physical consistency by respecting the geometric properties of the Earth, thereby improving long-term stability on the sphere. These methods facilitate the learning of stable dynamics by operating in spectral spaces that naturally align with global fluid flow characteristics. Despite these advancements, challenges remain in balancing the computational cost of evaluating complex physical residuals against training efficiency. Consequently, hybrid strategies that combine differentiable physics-

based parameterizations with deep learning backbones offer a promising pathway to achieve both high predictive skill and strict physical realizability in operational forecasting systems.

4.1.2 Hybrid Nudging and Spectral Blending Techniques

Hybrid nudging and spectral blending techniques have emerged as critical methodologies for integrating data-driven forecasts with physics-based constraints, addressing the inherent spectral biases of pure machine learning models. While neural operators excel at capturing large-scale synoptic patterns, they frequently exhibit energy dissipation at high wavenumbers, leading to overly smooth fields that lack fine-scale structural coherence. Spectral blending mitigates this by decomposing forecast fields into frequency domains, where low-wavenumber components are retained from the AI prediction to preserve learned non-linear dynamics, while high-wavenumber details are injected from numerical weather prediction (NWP) analyses or stochastic generators to restore realistic turbulence and gradient sharpness.

The implementation of these hybrids often involves a dynamic nudging strategy that operates directly within the spectral space, ensuring a seamless transition between scales without introducing non-physical discontinuities. By applying scale-dependent weighting functions, these techniques effectively correct the underestimation of spectral power observed in models like FuXi and GraphCast at smaller spatial scales. This approach not only enhances the visual fidelity of variables such as precipitation and temperature but also improves the statistical reliability of ensemble spreads, ensuring that the variance across ensemble members accurately reflects uncertainty at both synoptic and mesoscale levels.

Furthermore, advanced iterations of these methods incorporate adaptive filtering thresholds that respond to local flow characteristics, allowing for more precise preservation of extreme events and coherent structures. Unlike static blending ratios, adaptive schemes adjust the cutoff wavenumber based on the specific meteorological regime, thereby optimizing the balance between the computational efficiency of AI inference and the physical consistency of NWP dynamics. This synergy enables hybrid systems to maintain the long-range dependency modeling capabilities of deep learning while adhering to the conservation laws and

energy cascades dictated by atmospheric physics, ultimately yielding forecasts with superior skill scores for both medium-range predictions and extreme event attribution.

4.1.3 Conservation Law Enforcement in Generative Models

Generative models in weather forecasting, including Variational Autoencoders and Diffusion Models, often struggle to inherently satisfy fundamental physical conservation laws such as mass and energy continuity. Unlike traditional numerical weather prediction systems that explicitly solve discretized partial differential equations, data-driven approaches typically minimize statistical error metrics like mean squared error. This objective frequently leads to over-smoothed predictions and a failure to preserve the integral constraints of the atmosphere, resulting in physically inconsistent trajectories that drift significantly over long lead times or during extreme event simulations.

To mitigate these discrepancies, recent methodologies have shifted towards embedding physical priors directly into the generative architecture or training objectives. Techniques include modifying loss functions to penalize violations of conservation equations, thereby forcing the latent space to respect thermodynamic constraints. Furthermore, hybrid frameworks are emerging that couple generative emulators with dynamical cores, using the neural network to model sub-grid processes while the numerical solver enforces global conservation. This integration aims to retain the computational efficiency of deep learning while ensuring that generated ensembles remain within the bounds of physical plausibility.

Despite these advances, enforcing strict conservation in high-dimensional, stochastic generative systems remains an open challenge, particularly for rare extreme events where training data is scarce. Current research indicates that while soft constraints improve general fidelity, they may not fully prevent the accumulation of non-physical artifacts during autoregressive rollout. Future directions emphasize the development of hard-constrained architectures, such as symplectic integrators within neural operators, to guarantee that every sampled trajectory adheres to the underlying conservation laws without compromising the diversity required for robust uncertainty quantification.

4.2 Perturbation Methods for Uncertainty Quantification

4.2.1 Initial Condition Perturbations and Ensemble Spreading

Initial condition perturbations serve as the primary mechanism for quantifying forecast uncertainty in both numerical weather prediction and emerging data-driven frameworks. Traditional approaches typically sample perturbations from Gaussian distributions applied to atmospheric variables or spherical harmonic coefficients, aiming to represent analysis errors while maintaining dynamical balance. However, standard global root-mean-square error cost functions often penalize extreme values, leading to overly smooth forecast outputs that systematically underestimate the intensity of tropical cyclones and heat waves. This smoothing effect subsequently dampens the surrounding atmospheric systems, limiting the fidelity of extreme event predictions.

A critical challenge in modern ensemble forecasting is the accurate calibration of ensemble spread, which frequently suffers from underdispersion, particularly in machine learning models. Recent investigations reveal that while Gaussian perturbations are computationally efficient, they often fail to capture the correct scaling of uncertainty growth, resulting in a "no butterfly effect" scenario where error growth for small perturbations remains insufficient. Although advanced architectures attempt to address this by integrating stochastic noise, such as Perlin noise at multiple spatial scales, many models still struggle to align their spread maxima with operational benchmarks, often capturing large-scale gradients while underestimating the amplitude of local variability.

To mitigate these deficiencies, contemporary research explores hybrid strategies that combine controlled initial perturbations with generative diffusion processes. These methods aim to reproduce the statistical properties of huge ensembles without the prohibitive computational cost of running hundreds of deterministic trajectories. By adjusting perturbation magnitudes to match the temporal evolution of standard deviations observed in general circulation models, these approaches seek to stabilize forecast amplitudes and improve hazard estimates. Nevertheless, achieving a spatially regular pattern in ensemble spread that accurately reflects the non-linear dynamics of extreme events remains a non-trivial objective, requiring careful

tuning of noise injection mechanisms to avoid irregular artifacts.

4.2.2 Adversarial Attacks and Model Robustness Analysis

Adversarial attacks in AI-driven weather prediction exploit the high-dimensional sensitivity of neural networks to imperceptible input perturbations, potentially leading to catastrophic forecast failures. Unlike traditional numerical models, deep learning architectures often lack inherent physical constraints, making them susceptible to adversarial examples that shift predictions away from realistic atmospheric states. Research indicates that standard deterministic models can exhibit significant error growth when subjected to targeted noise, particularly in regions characterized by chaotic dynamics or extreme events, thereby undermining operational reliability.

To mitigate these vulnerabilities, robustness analysis increasingly employs adversarial training strategies and physics-informed regularization techniques. Methods such as domain-adversarial training and the integration of bred vector perturbations aim to align model gradients with physically consistent trajectories, reducing the likelihood of non-physical outputs. Furthermore, generative approaches, including Generative Adversarial Networks and diffusion models, are utilized to synthesize extreme scenarios, forcing the predictor to learn robust representations of rare events rather than merely minimizing mean squared error on average conditions.

Despite these advances, a critical challenge remains in balancing model sensitivity with stability, often manifesting as a bias-variance trade-off where excessive smoothing suppresses extreme values. Current evaluation frameworks frequently rely on limited case studies, failing to capture the full spectrum of out-of-distribution behaviors encountered in real-world deployment. Future robustness assessments must therefore incorporate rigorous stress testing against synthesized adversarial patterns and ensure that uncertainty quantification methods accurately reflect both aleatoric and epistemic sources of error under perturbed initial conditions.

4.2.3 Rare Event Sampling via Large Deviation Theory

Large Deviation Theory (LDT) provides a rigorous mathematical framework for quantifying

the exponentially small probabilities of rare atmospheric events, addressing the inherent data scarcity that limits pure AI models. By characterizing the rate functions governing tail distributions, LDT enables the systematic generation of extreme trajectories that are statistically consistent with underlying physical dynamics yet absent in standard observational datasets. This theoretical foundation is critical for overcoming the excessive smoothing typical of deep learning architectures, which often fail to capture the sharp gradients and high-amplitude anomalies characteristic of severe weather phenomena.

Integrating LDT with rare event sampling algorithms allows for the efficient exploration of phase space regions associated with extreme outcomes, such as prolonged heatwaves or rapid cyclone intensification. Unlike brute-force Monte Carlo simulations, which are computationally prohibitive for return periods exceeding decades, LDT-guided methods utilize importance sampling or cloning strategies to bias trajectories toward critical instants. These techniques effectively amplify the occurrence of low-probability states during training, ensuring that AI models encounter a balanced distribution of extreme scenarios and learn the non-linear mechanisms driving their evolution without relying solely on sparse historical records.

The application of these sampling strategies significantly enhances the generalization capabilities of AI weather prediction systems regarding tail events. By augmenting training datasets with synthetically generated extremes derived from large deviation principles, models can better estimate conditional quantiles and reduce amplitude underestimation in probabilistic forecasts. This approach not only improves the reliability of predicting threshold exceedances but also ensures that the learned representations adhere to physical constraints, thereby mitigating the risk of generating spurious anomalies while preserving the statistical fidelity required for robust emergency management and climate risk assessment.

4.3 Attribution and Interpretability of Extreme Events

4.3.1 Counterfactual Simulations for Anthropogenic Attribution

Counterfactual simulations have emerged as a pivotal methodology for attributing specific extreme weather events to anthropogenic climate change,

addressing the limitations of traditional statistical approaches that often suffer from selection bias and computational inefficiency. Unlike conventional methods that rely solely on historical observational records or massive ensemble runs of physics-based global climate models, counterfactual frameworks construct alternative realities where anthropogenic warming signals are explicitly removed. This involves detrending sea surface temperatures and atmospheric variables to isolate the influence of human activity, thereby enabling a direct comparison between the observed event and its hypothetical natural counterpart. By generating these "worlds that might have been," researchers can quantify changes in event probability and intensity with greater physical interpretability than pure extreme value theory extrapolations.

The integration of artificial intelligence weather prediction models into this workflow offers a transformative advantage in computational speed and scalability, facilitating the generation of large ensembles required to sample rare, high-impact excursions far from normal system behavior. While numerical weather prediction models strictly solve atmospheric governing equations, data-driven counterparts learn fundamental physical relationships from reanalysis data, allowing for rapid simulation of weather evolution across varying lead times. However, the application of these deterministic AI models to attribution studies necessitates rigorous verification to ensure they accurately capture the underlying flow dynamics governing extreme events rather than merely reproducing climatological means. Challenges remain regarding the sensitivity of these models to initial conditions and their ability to reproduce the chaotic butterfly effect essential for distinguishing between natural variability and forced climate responses.

To ensure robust attribution outcomes, recent frameworks prioritize event-based experimental designs that define extreme occurrences through specific criteria regarding onset, peak, and recovery phases using high-quality reanalysis datasets. These approaches often employ hybrid strategies, combining the efficiency of AI-driven simulations with the physical constraints of traditional models to assess risks associated with heatwaves, floods, and other disasters. By utilizing open-source tools and accessible data, the scientific community aims to enhance replicability and reduce the latency between event occurrence and attribution analysis.

Ultimately, refining these counterfactual techniques is critical for informing public health responses and adaptation strategies, particularly as the frequency and severity of anthropogenic extremes continue to escalate across diverse geographic regions.

4.3.2 Explainable AI for Feature Importance and Causality

Explainable AI (XAI) has become indispensable for diagnosing the physical fidelity of data-driven weather prediction models, particularly when addressing failures in extreme event forecasting such as sudden-turning typhoons. Traditional black-box architectures often exhibit excessive smoothing of intensity peaks due to restricted sample sizes and a lack of explicit physical constraints. By deploying feature importance techniques like Layer-Wise Relevance Propagation, researchers can quantify the contribution of specific input modalities, revealing that while wind components and pressure gradients dominate short-term predictions, temporal features often drive longer horizons. This granular attribution allows scientists to verify whether models rely on genuine atmospheric dynamics or spurious statistical correlations, thereby identifying regimes where pure AI approaches diverge from established numerical weather prediction standards.

Beyond static feature ranking, emerging methodologies aim to uncover causal mechanisms within latent spaces to distinguish between mere association and physical causality. Sensitivity analyses applied to specific cyclones demonstrate that advanced models can implicitly learn governing equations, such as pressure-gradient-driven fluid motion, yet they frequently struggle with out-of-distribution events like unprecedented rainfall or cold surges. Integrating causal discovery frameworks enables the isolation of subseasonal drivers and the assessment of how forecast skill evolves across different physical processes. This mechanistic understanding is critical for determining if a model's success stems from learning robust atmospheric physics or overfitting to historical reanalysis patterns, which is essential for reliable attribution studies of human-induced climate change impacts.

To bridge the gap between predictive accuracy and physical interpretability, future research must prioritize hybrid architectures that embed explicit conservation laws alongside data-driven learning.

Current benchmarks indicate that while AI models offer superior computational efficiency, they require rigorous oversight to prevent underestimation of record-breaking intensities. Explainable workflows facilitate this by providing event-wise predictability scores and visual analytics that highlight discrepancies between AI emulators and operational ensembles. Ultimately, fostering trust in AI-driven early warning systems depends on transitioning from post-hoc explanations to inherently interpretable models that can robustly generalize across diverse climatic regimes and provide actionable insights for high-stakes decision-making.

4.3.3 Dynamical Precursor Identification and Pattern Recognition

Dynamical precursor identification leverages analog-based methods to reconstruct synthetic time series by matching current atmospheric states with historical circulation patterns exhibiting similar thermodynamic characteristics. By constructing catalogs of analogs from reanalysis datasets like ERA5, algorithms compute sets of nearest neighbors based on specific metrics constrained by calendar windows to ensure seasonal consistency. This approach allows for the extraction of dynamical history, where the evolution of a selected analog after a specific lag time serves as the subsequent member of a synthetic trajectory, effectively capturing the non-linear progression of extreme events without relying solely on deterministic model integration.

Advanced pattern recognition techniques further enhance predictability by isolating coherent structures and synoptic regimes that precede high-impact weather phenomena. Convolutional neural networks and transformer architectures are increasingly employed to diagnose extreme heat precursors and track the spatial migration of high-pressure anomalies weeks in advance. These models analyze geopotential height fields to identify persistent quadrupoles and strain cores, distinguishing between highly predictable extremes characterized by long-term structural persistence and those with limited forecast horizons due to rapid dynamical decorrelation.

The integration of these identification strategies into early warning systems addresses fundamental predictability constraints by focusing on event-wise, skill-based diagnostics rather than aggregate statistical performance. By quantifying the temporal persistence of specific flow configurations, re-

searchers can establish precise predictability limits directly from observational data. This methodology not only improves the detection of rare events through representative catalog generation but also provides a robust framework for understanding the mechanistic drivers of extreme weather, bridging the gap between traditional dynamical systems theory and modern data-driven forecasting capabilities.

5 Future Directions

Despite significant advancements in deep learning for extreme weather forecasting, several critical limitations persist that hinder operational deployment. Current data-driven models often struggle with data scarcity regarding rare, high-impact events, leading to poor generalization in the statistical tails of atmospheric distributions. Generative approaches, while promising for ensemble creation, frequently suffer from mode collapse or physical inconsistencies, producing scenarios that violate fundamental conservation laws or lack temporal coherence over extended lead times. Furthermore, the integration of physical constraints remains largely superficial, with many architectures functioning as black boxes that offer limited interpretability regarding the dynamical drivers of extreme phenomena. The disconnect between global modeling capabilities and the need for localized, kilometer-scale resolution also presents a persistent challenge, particularly when downscaling coarse outputs without introducing artifacts or losing fine-scale storm dynamics essential for impact assessment.

Future research must prioritize the development of robust hybrid frameworks that deeply embed physical laws directly into neural architectures rather than treating them as soft regularization terms. This involves creating physics-informed generative models where conservation of mass, energy, and momentum are hard constraints within the network topology, ensuring that simulated extremes remain dynamically plausible. Concurrently, there is an urgent need to advance self-supervised and few-shot learning techniques to effectively leverage sparse historical records of rare events, enabling models to learn robust representations of extremes without relying solely on massive labeled datasets. Additionally, future work should focus on enhancing the interpretability of these systems through neuro-symbolic approaches

that can explicitly articulate the causal mechanisms driving predicted events, thereby fostering trust among meteorologists and decision-makers. The integration of multimodal data sources, including satellite imagery, radar observations, and unstructured textual reports, via advanced vision-language models should also be expanded to create more comprehensive situational awareness systems.

The successful realization of these research directions holds transformative potential for global societal resilience against climate change. By delivering forecasts that are not only computationally efficient but also physically consistent and reliable in predicting rare extremes, next-generation models will significantly improve early warning systems for tropical cyclones, heatwaves, and flash floods. Enhanced uncertainty quantification through high-fidelity generative ensembles will empower policymakers and emergency responders to make more informed, risk-aware decisions, ultimately reducing economic losses and saving lives. Moreover, bridging the gap between artificial intelligence efficiency and numerical weather prediction fidelity will democratize access to high-resolution forecasting capabilities, allowing developing nations to better prepare for escalating weather hazards and adapt to a changing climate landscape.

6 Conclusion

This survey has systematically examined the transformative role of deep learning in forecasting extreme weather events, highlighting a distinct shift from traditional numerical methods to data-driven architectures capable of capturing non-linear atmospheric dynamics. The review established that global modeling has been revolutionized by transformer and graph-based networks, which effectively handle the non-Euclidean geometry of the Earth to preserve long-range teleconnections essential for medium-range predictions. Concurrently, localized nowcasting has advanced through hybrid convolutional and recurrent frameworks that resolve fine-scale storm structures with unprecedented temporal fidelity. Furthermore, the integration of generative approaches, including diffusion models and Generative Adversarial Networks, has addressed the critical challenge of uncertainty quantification. These models successfully overcome the smoothing biases inherent in

deterministic regression, enabling the generation of high-resolution ensemble members that accurately represent the statistical tails of rare, high-impact phenomena such as tropical cyclones and severe precipitation.

The significance of this synthesis lies in its unified taxonomy, which bridges the gap between theoretical architectural innovations and practical operational requirements for hazard mitigation. By critically analyzing the convergence of geometric deep learning, attention mechanisms, and probabilistic generative frameworks, this work elucidates how modern artificial intelligence can surpass the computational latency and resolution limits of conventional Numerical Weather Prediction systems. The survey underscores the necessity of hybrid physical-data frameworks that embed conservation laws and causal constraints directly into neural objectives, ensuring that predictions remain not only statistically accurate but also physically consistent. This comprehensive perspective provides a foundational roadmap for developing next-generation forecasting systems that combine the efficiency of machine learning with the fidelity of physical laws, thereby enhancing societal resilience against escalating climate extremes.

Looking forward, the meteorological community must prioritize the development of robust hybrid models that mitigate current limitations such as mode collapse in generative networks and data scarcity for rare events. Future research should focus on refining physics-augmented learning objectives to ensure interpretability and reliability in operational settings, moving beyond black-box predictions toward transparent, trustworthy systems. There is an urgent call to action for interdisciplinary collaboration between computer scientists, meteorologists, and climate physicists to create standardized benchmarks and open datasets specifically tailored for extreme event modeling. By fostering this synergy, the field can accelerate the deployment of AI-driven tools that provide timely, accurate, and actionable insights, ultimately safeguarding global infrastructure and human life against the intensifying threats of a changing climate.

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